

Course Code: BET-C306
Course Name: DIGITAL SYSTEM DESIGN

MM: 100 Time: 3 Hr. L T P 3 0 0	Sessional: 30 ESE: 70 Credit : 3
--	--

Prerequisites:	For this course, no pre-requisites are required. But should have knowledge of Diodes, transistors.
Objectives:	1. To get good knowledge of digital system. 2. Learn about the different number system that have different bases which plays very significant role in computer world. 3. During the course we can learn how to design the digital circuits by using Boolean algebra, K-maps and logic gates. 4. And to enable to implement synchronous state machines using flip-flops.
Course Coordinator	Dr. Tanuj Kumar Garg

NOTE:	The question paper shall consist of two sections (Section-A and Section-B). Section-A shall contain Ten (10) short answer type questions of six (06) mark each and student shall be required to attempt any five (05) questions. Section-B shall contain eight (08) long answer type questions of ten (10) marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus
--------------	--

UNIT	Module	Course Content	No. of Hours	POs mapped	PSOs mapped
UNIT-1	Module-1	Number System: Representation of negative numbers, 9's and 1's complement, 10's and 2's complement, arithmetic using 2's complement. BCD Code, Gray Code, Excess-3 Code, Introduction to Boolean algebra, Truth table verification of various gates, Realization of Switching functions with gates.	3	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2
	Module-2	K- Map: Representation up to 4 variables, simplification and realization of various functions using gates, Tabular Method, Combinational logic and design procedure.	3	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2
UNIT-2	Module-3	Combinational logic Circuits: Arithmetic circuits, Half and Full adder, Subtractors, BCD adders, Code Conversion, 4 bit Magnitude Comparator (IC -7485), Cascading of IC 7485, Decoder, Multiplexer, Demultiplexers, Encoders. Parallel Binary adder, IC 7483, 4-bit Binary parallel adder/subtractor,	9	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2
UNIT-3	Module-4	Sequential Logic Circuits: Flip Flops, S-R latch, gated latches, Edge triggered Flip Flops, Master-slave Flip Flops, Conversion of flip flops, Analysis of clocked sequential circuits, Design of synchronous circuits, State transition diagram, state reduction and assignment.	10	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2

UNIT-4	Module-5	Counters: Design of Asynchronous and Synchronous Counters, Two bits & four bits up & down counters and their design, Shift registers, Serial & Parallel data transfer, Shift left/Right register, Shift Register applications.	8	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2
UNIT-5	Module-6	Logic Families: Diode switching, Transistors as a switching element, MOS as a digital circuit element, concept of transfer characteristics, input characteristics and output characteristics of logic gates, fan in, fan out, noise margin, Logic families: TTL, IIL, ECL, NMOS, & CMOS, Open collector outputs.	7	PO1/ PO2/ PO3/ PO5/ PO10/PO12	PSO1/ PSO2
Total No. of Hours			40		

Learning Outcomes:	At the end of this course students will demonstrate the ability to • Design and analyze combinational logic circuits. • Design & analyze modular combinational circuits with MUX/DEMUX, Decoder, Encoder . • Design & analyze synchronous sequential logic circuits.
---------------------------	--

Suggested books:

S. No.	Name of Authors /Books /Publisher
1.	M.Morris Mano, Digital Design, PHI
2.	R.P.Jain, Modern Digital electronics, TMH
3.	A.Anand Kumar, Fundamentals of Digital Circuits, PHI
4.	Lee S.C, Modern Switching Theory and Digital design, PHI
5.	Greenfield J.D., Practical Digital design using ICs, John Wiley.

	CO-PO/PSO MAPPING													
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3		3	3	3							3	3	3
CO2	3		3	3	3							3	3	3
CO3	3		3	3	3							3	3	3
CO4	3		2	1	3							3	3	3
CO5	3		2	1	3							3	3	3
CO6	3		1	1	3							3	3	3
CO7	3		1	3	3							3	3	3
CO8	3		1	3	3							3	3	3
CO9	3		1	3	3							3	3	3
CO10	3		1	1	3							3	3	3